MIL-Y-3388B

28 March 1966
SUPERSEDING
MIL-Y-3388A
27 June 1962

MILITARY SPECIFICATION

YEAST FOOD, MINERAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a mineral yeast food for use in the production of yeast-raised baked products in the kitchens and bakeries of the Armed Forces.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-S-48	- Sacks, Shipping, Paper.
PPP-B-35	- Bags, Textile, Shipping, Burlap.

MILITARY

MIL-I-21330	- Insect Resistant Treatment for Paper.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-668	- Sanitary Standards for Food Plants.

FSC 8950

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:36Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply:

U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

Association of Official Agricultural Chemists

Official Methods of Analysis

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D. C. 20044.)

U. S. Pharmacopoeial Convention, Inc.

Pharmacopoeia of the United States

(Application for copies should be addressed to the U. S. Pharmacopoeial Convention, Inc., 46 Park Ave., New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Preproduction sample approval. - When specified (see 6.1), a sample of the finished product shall be submitted to the contracting officer for approval before production is commenced.

3.2 Material. - All material shall be of edible grade and shall be clean, sound, and wholesome. They shall be free from all evidence of insect infestation or other objectionable foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:36Z
Check the source to verify that this is the current version before use.

3.2.1 Ammonium chloride, calcium sulfate, and potassium bromate. - These ingredients shall comply with the purity standards of U. S. Pharmacopoeia grade.

3.2.2 Salt. - Salt shall be white, refined, and of the style known in the trade as "Flour style." Iodized salt shall not be used.

3.2.3 Cornstarch. - Cornstarch shall be pure, unmodified cornstarch. It shall have a natural starch flavor and odor, and shall be free from rancid, musty, sour, and other undesirable flavors and odors.

3.2.4 Wheat flour. - Flour shall be free from musty, sour, garlicky, and all other flavors and odors not characteristic of freshly milled, sound, clean wheat. The flour shall be free from all evidence of insect infestation. The flour may be bleached or unbleached.

3.3 Preparation. -

3.3.1 Formula. - The product shall contain the following ingredients within the limits of the designated percent by weight:

Material	Percent by weight (as is) Minimum	Maximum
Ammonium chloride	9.3	9.7
Calcium sulfate	24.5	42.2
Potassium bromate	0.26	0.30
Salt (Sodium chloride)	24.5	36.8
Cornstarch or wheat flour	Balance to total 100 percent.	

3.4 Finished product. - The finished product shall meet the requirements in 3.3.1. It shall have a moisture content of not more than 11.0 percent by weight. It shall be free from lumps which do not break up under light pressure.

3.5 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement.

3.6 Workmanship. - The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared and processed with minimum delay between the various stages in production.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:36Z
Check the source to verify that this is the current version before use.

of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection (plant sanitation inspection). - Product shall be rejected if produced in plants which fail to meet sanitary requirements of MIL-STD-688.

4.3 Sampling for inspection. - Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.3.1 Component and material inspection. - In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of ingredients. - Conformance of ammonium chloride, calcium sulfate, potassium bromate and salt, to 3.2 through 3.2.2, as concerns identity (kind, grade, and style, as applicable), shall be accomplished by review of invoices, labels or other valid documents. In addition, wheat flour or cornstarch, as applicable, shall be examined organoleptically as necessary to determine compliance with applicable requirements of 3.2.3 or 3.2.4. The sample unit shall be 1 pound derived from each of 5 primary containers (or all containers if less than 5 form a lot). Any nonconformance shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved quantity of finished product.

4.3.1.2 Insect-resistant treated sacks. - Conformance to 5.2.1 as concerns the requirement for insect-resistant treatment and date of fabrication shall be ascertained by examination of markings on shipping sacks. Use of nonconforming shipping sacks shall be cause for rejection of the involved end item.

4.3.2 Examination of end item. - Examination shall be in accordance with tables II and III and 4.3.2.1 and 4.3.2.2. The sampling plans and acceptance criteria shall be as shown in table I. Acceptable quality levels (AQLs) shall be expressed in terms of defects per hundred units:

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:36Z
Check the source to verify that this is the current version before use.

TABLE I - Sampling plans and AQLs.

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
II	S-3	1 primary container	Primary containers	---	2.5
III	S-2	1 pound from a primary container	Primary containers	1.5	---

4.3.2.1 Examination of primary container. - Examination of bags shall be in accordance with the criteria for examination of closure contained in PPP-B-35. In addition, the following defects shall be added to the table of examination: Major - Tear or hole. Minor - Not clean.

TABLE II - Examination for net weight. 1/

Category	Defect
Minor	
201	Net weight less than 49-1/2 pounds.

1/ Report results to nearest 1/4 pound. Lot shall be rejected if sample data indicate lot average net weight less than 50 pounds.

TABLE III - Examination of finished product. 1/

Category	Defect
Major	
152	Not free from lumps which do not break up under light pressure.

1/ Presence of foreign material, such as but not limited to, dirt, glass, insect, insect fragment, paint, metal, smut, ergot, weed seeds, wood particles, animal excreta, etc., shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:36Z
Check the source to verify that this is the current version before use.

4.3.2.2 Examination of shipping sacks. - When multiwall paper shipping sacks are required to be in accordance with UU-S-48, examination of the filled and sealed shipping sacks shall be in accordance with the external examination of the end item (except as pertains to count). The sample size shall be indicated by inspection level S-3, and the AQL shall be 4.0, expressed in terms of defects per hundred units, for each table of examination.

4.3.2.3 Examination for marking of level C of shipping containers. - Examination shall be as follows: The sample unit shall be one shipping container. Lot size shall be expressed in terms of shipping containers, and the sample size shall be the number of shipping containers indicated by inspection level S-3. A defect shall be scored for each shipping container with marking missing, incomplete or not as specified. The AQL, expressed in terms of percent defective, shall be 2.5.

4.3.3 Sampling procedure and acceptance criteria for testing of finished product. - Procedures for testing for ammonium chloride, calcium sulfate, potassium bromate, salt and moisture content, shall be in accordance with 4.5.1 and 4.5.1.1. Test requirements shall be lot average requirements. The sample for testing shall be a 1-pound composite derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of primary containers. Results shall be reported to the same decimal place as specified in the requirements. Any test failure shall be cause for rejection of the lot.

4.4 Standby test samples. - The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier.

4.5 Tests. - Test procedures and controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.5.1 Chemical analyses. - Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Agricultural Chemists (AOAC). All tests, except moisture and Potassium Bromate, shall be made in accordance with chapter: Baking Powders and Baking Chemicals of the AOAC. Moisture shall be determined in accordance with chapter: Cereal Foods, section: Wheat Flour. Test for potassium bromate shall be made in accordance with 4.5.1.1.

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:36Z
Check the source to verify that this is the current version before use.

Test	Method
Moisture	Vacuum oven
Calcium sulfate	Calcium
Ammonium chloride	Ammonia
Salt (sodium chloride)	Sodium

4.5.1.1 Test for potassium bromate. - Test shall be performed as follows: Weigh 50.00 grams of mineral yeast food into a clean, dry, 800-ml. beaker. Add a tablespoon of gypsum (or some filter-aid) and 500 ml. of water. Stir with a glass rod to break up caked portion on bottom of beaker, and then with a motor until all soluble salts are in solution (3 to 4 minutes). Filter a portion of the suspension, using suction, with a 500-ml. filter flask, a Buechner funnel, and No. 1 Whatman filter paper. Transfer a 50-ml. aliquot of the filtrate (equivalent to 4.902 grams of sample) to a 250-ml. glass-stoppered flask, add 3 grams of potassium iodide, followed by 3 ml. of hydrochloric acid. Stopper the flask and allow to stand 5 minutes; add 100 ml. of cold water, and titrate the liberated iodine with tenth-normal sodium thiosulfate, using starch solution as indicator. Correct for a blank, made with the same quantities of the same reagents. The following formula shall be used for calculating the percent of potassium bromate: (ml. N/10 sodium thiosulfate) x 0.0568 = percent $KBrO_3$.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - The product shall be packaged in accordance with 5.1.1 or 5.1.2, as specified (see 6.1).

5.1.1 Level A. - Fifty pounds of the product shall be packaged in a textile bag complying with type IV of PPP-B-35, as specified for the commodity listing for "Flour".

5.1.2 Level C. - Fifty pounds of the product shall be packaged directly into the shipping container specified in 5.2.2.

5.2 Packing. - The product shall be packed in accordance with 5.2.1 or 5.2.2, as specified (see 6.1).

5.2.1 Level A. - One 50-pound bag of the product shall be packed in a multi-wall paper shipping sack complying with type II, shipping sack No. 31X of UU-S-48. The ends of the shipping sack shall be wax dipped. The shipping sack shall have the outer ply insect-resistant treated (IRT) in accordance with MIL-I-21330. The exterior surface of each completed sack shall contain the following statement in letters not less than 1/2-inch high:

IRT - SACK FABRICATED _____
(day, month, year)

The IRT sack shall be fabricated not more than 60 days prior to shipment of the product.

Source: https://assist.dla.mil -- Downloaded: 202640-16T17:36Z
Check the source to verify that this is the currentversion before use.

5.2.2 Level C. - Fifty pounds of the product shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the rules or regulations of carrier applicable to the mode of transportation.

5.3 Labeling and marking. -

5.3.1 Bags. - Labeling is not required.

5.3.2 Shipping containers. - Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether preproduction sample is to be furnished (see 3.1).
(c) Levels of packaging and packing required (see 5.1 and 5.2).

6.2 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 67

Preparing activity:

Army - GL

Project No. 8950-0063

Review activities:

Army - MD
Navy - MC, SA
Air Force - 67

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Documents.

Source: https://assist.dla.mil -- Downloaded: 20260-16T17:36Z
Check the source to verify that this is the current version before use.